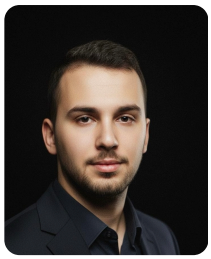
Muhammet Utku Küçük

Backend Software Engineer

Istanbul, Türkiye | +90 554 006 75 23 | muhammetutku@outlook.com | linkedin.com/in/utkukucuk | github.com/utku-kucuk



SUMMARY

Backend Software Engineer building, deploying and operating RESTful services and Linux-based infrastructure across corporate, remote and R&D roles. Focused on **Java and Spring Boot**, with additional experience in Python, Go and PHP. Delivered Web3 data integrations, a video-streaming platform and IoT backends, and containerized **10+ services** with Docker, cutting deployment time from ~2 hours to ~20 minutes. Experienced with API security (JWT, RBAC), PostgreSQL and MongoDB data modeling, Nginx and CI/CD with GitHub Actions. Aiming to grow toward cloud-native architecture with Kubernetes and AWS.

TECHNICAL SKILLS

Languages: Java, Python, Go, C#, SQL, PHP, JavaScript, TypeScript

Backend: Spring Boot, Spring Security, Spring Data JPA, Hibernate, RESTful APIs, Microservices, JWT, RBAC, OpenAPI/Swagger, Node.js, Express.js

Databases: PostgreSQL, MySQL, MSSQL, MongoDB, SQLite, Realm (offline-first)

Cloud & DevOps: Docker, Docker Compose, Linux, Nginx (reverse proxy, SSL/TLS), GitHub Actions (CI/CD), AWS (EC2, S3)

Testing & Tools: JUnit, Mockito, Postman, Git, Maven, Agile/Scrum, Code Review

IoT: C/C++ (embedded), ESP32, ESPHome, MQTT, Home Assistant

EXPERIENCE

R&D Project Evaluator | Zaim Teknopark · Freelance · Remote — Oct 2026 – Present

- Perform technical due diligence on R&D project proposals from technology companies, assessing feasibility, innovation level and R&D eligibility against Teknopark evaluation criteria.
- Assess the architecture, technology stack and delivery plans of backend, cloud and IoT projects; audit progress and final reports and issue approval or revision recommendations.

Backend Developer | Covalent (CQT) · Contract · Remote — Sep 2025 – Feb 2026

- Built backend services on top of the Covalent Unified API, delivering on-chain data (token balances, transactions, NFT metadata) from **10+ blockchain networks** to client applications.
- Developed a token/NFT marketplace dApp end-to-end: REST API layer, wallet-based authentication, listing and transaction-history endpoints, and smart-contract interaction.
- Introduced response caching and request batching, cutting external API calls by **~40%** and average endpoint latency by **~35%**.
- Hardened integrations with pagination, input validation and rate-limit handling (retry with exponential backoff); documented APIs with OpenAPI and shipped in agile sprints with PR-based code reviews.

Software Developer – Computer Vision | Nova UAV Team · University R&D Team · Istanbul — Nov 2024 – Apr 2025

- Developed a real-time image-processing pipeline in Python and OpenCV for UAV camera feeds, covering frame preprocessing, color segmentation and object detection.
- Optimized frame-processing logic, raising throughput from **~15 to ~25 FPS** on onboard hardware.

Software Specialist – Backend & DevOps | Teknolojik Çözüm · Istanbul — Aug 2022 – Apr 2023

- Built PHP and JavaScript backend services, REST APIs and MySQL-backed admin panels for **5+ corporate clients**.
- Containerized **10+ services** with Docker and Docker Compose, standardizing environments and reducing deployment time from **~2 hours to ~20 minutes**.
- Configured Nginx as a reverse proxy with SSL/TLS and routing to securely expose services on Linux servers; set up CI/CD pipelines with GitHub Actions.
- Drove log analysis and root-cause troubleshooting in production, reducing recurring incidents by **~30%** and sustaining **99.5%+ uptime**.

Software Development Intern | BİL Holding · Istanbul Oct 2020 – Jan 2021

- Built responsive UI components with HTML, CSS, JavaScript and Bootstrap and integrated them with backend REST APIs; supported mobile and Arduino-based prototypes.

Software Development Intern | Kale Kalıp · Istanbul Mar 2020 – May 2020

- Implemented and unit-tested business-logic modules for internal applications; collected and processed sensor data from production-floor IoT devices.

PROJECTS

Commerce Platform github.com/utku-kucuk/commerce-platform

Java, Spring Boot 3, Microservices, Spring Security, Spring Data JPA, PostgreSQL, Docker, JUnit

- E-commerce backend built as Spring Boot microservices (catalog, cart and order domains) exposing RESTful APIs documented with OpenAPI.
- Stateless JWT authentication and role-based access control with Spring Security; unit and integration tests with JUnit and Mockito; services containerized with Docker and PostgreSQL.

Viducate – Online Video Learning Platform github.com/utku-kucuk/viducate

Node.js, Express.js, PostgreSQL, FFmpeg, Docker, Nginx

- Designed the video pipeline: chunked, resumable uploads for multi-GB lecture videos, FFmpeg transcoding to HLS adaptive streams and PostgreSQL-backed metadata.
- Implemented JWT authentication with three-role RBAC (Admin, Teacher, Student); deployed on Linux with Docker behind an Nginx reverse proxy.

IronEcho – Self-Hosted Fitness Platform github.com/utku-kucuk/IronEcho

Node.js, TypeScript, Express.js, MongoDB, Realm, Docker Compose

- Designed an offline-first backend with custom data-synchronization and conflict-resolution logic, JWT authentication and MongoDB data models for workouts and analytics; shipped as a one-command Docker Compose deployment.

Smart Home IoT Suite github.com/utku-kucuk/SmartBlind-ESP32

ESP32, ESPHome, MQTT, Home Assistant, Docker, Linux

- ESP32 devices (smart blind, smart lamp, climate sensors) integrated over MQTT into a self-hosted Home Assistant server.

EDUCATION

B.Sc. in Computer Engineering (English) | Istanbul Aydın University 2022 – 2025

Started at Bahçeşehir University, Computer Engineering (2020 – 2022); completed via lateral transfer.

Associate Degree in Computer Programming | Istanbul Aydın University 2018 – 2020

Information Technologies (Software & Networking) | Avcılar Anatolian Technical High School 2014 – 2018

Vocational training in programming and computer networks; IT internships at Kale Holding and Boğaziçi Döküman Çözümleri.

LANGUAGES

Turkish (Native) · English (B2 – Professional Working Proficiency)

REFERENCES

Yasin Kara – Founder, Teknolojik Çözüm and Mahir Bilişim

Prof. Dr. Zafer Arslan – Department of Computer Engineering, Istanbul Aydın University

Serkan Seyhan – Co-Founder, Quantum Plasma Technology · Board Member, Bizim Bulut

Contact details available upon request.